

Supplementary Material

UAS data pre-processing

We collected all UAS images using Phantom 4 pro system with RGB camera using below specification (Supplementary Table 1). The initial processing of UAS images were done using agisoft metashape software following the specifications in Supplementary Table 1. After loading the UAS images, we first convert all the images into EPSG 26913 reference system. The conversion allows for direct comparison of data to in-situ measurements (individual seedling locations and their size) as well as to further process the point cloud to delineate individual trees. After coordinate conversion, we removed images collected on the ground and outside the flight boundary. Then image quality was estimated. Only the images with quality greater than 0.7 were used for further processing. The quality images were aligned using parameters mentioned in Supplementary Table 1. The tie points resulting from the aligned images were further refined using gradual selection parameters. The cleaned tie points were used to build the dense point cloud. The dense cloud is converted to a digital surface model (DSM). Finally the digital surface model was used to generate the orthomosaic. The final spatial resolution of the digital surface model was approximately 3 cm. After processing the images in Agisoft metashape we exported both the orthomosaic (as a tif file) and point cloud data (.laz file). These exported data were used in the R statistical program for further processing as described in methods of the main manuscript.

Supplementary Table 1: UAS Specifications and flight parameters specified in this study

Parameter Type	UAS Flight Parameter	Parameter Details
UAS Specifications	Vehicle	DJI Phantom 4 Pro
	Payload	Integrated off-the-shelf RGB camera
	Sensor	1 inch CMOS sensor, 20M effective pixels
	Lens	FOV 84° 8.8 mm/24 mm (35 mm format equivalent) f/2.8 - f/11 auto focus at 1 m - ∞
	ISO Range	100-3200 (Auto)
	Shutter	Electronic & Mechanical shutter
UAS Flight Parameters	Flight planning software	Map Pilot Pro
	Altitude	100m Above Ground Level (AGL)
	Along track overlap	85%
	Across track overlap	85%
	Flight speed	6.0 m/s
	Terrain follow use	Activated: Elevation profile USGS 30m SRTM
	Flight pattern	Grid mission

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Table 2: UAS data processing parameters in the Agisoft metashape software.

Parameter	Value
Coordinate system	NAD83 / UTM zone 13N (EPSG:26913)
Rotation angles	Yaw, Pitch, Roll
Point colors	3 bands, uint8
Dense Point Cloud	
Depth maps generation quality	High
Filtering mode	Aggressive
File size	~ 0.2 GB
DEM	
Source data	Dense cloud

Parameter	Value
Interpolation	Extrapolated
Software version	1.8.3.14331
Orthomosaic	
Colors	3 bands, uint8
Blending mode	Mosaic
Surface	DEM
Hole filling	Enabled
Ghosting filter	Disabled
Processing time	57 minutes 26 seconds
System	
Software name	Agisoft Metashape Professional
Software version	1.8.3 build 14331
OS	Linux 64-bit
RAM	250.98 GB
CPU	AMD EPYC 7402 24-Core Processor
GPU(s)	Quadro RTX 8000 (2x)

Vegetation indices, Texture, and point cloud based variables

We derived a set of red, Green, and Blue band indices as well as a set of texture variables describing pixel reflectance values within a given neighborhood (in our study it is the canopy crown area) from each band. In addition, we derived a few point cloud based variables. The point cloud based variables were derived using points within each delineated canopy polygon.

Table 3: Summary table of variables used to classify vegetation classes in the main study

Type	Name	Formula/description	Reference
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Bands	Red reflectance (R)	Reflectance of the red band	
	Green reflectance (G)	Reflectance of the green band	
	Blue reflectance (B)	Reflectance of the blue band	
Indices	Red index (r)	$r = R/(R + G + B)$	(Lu et al., 2019)
	Green index (g)	$g = G/(R + G + B)$	(Lu et al., 2019)
	Blue Index (b)	$b = B/(R + G + B)$	(Lu et al., 2019)
	Excess Red Vegetation Index (EXR)	$EXR = 1.4 r - g$	(Lu et al., 2019)
	Visible Atmospherically Resistant Index (VARI)	$VARI = (g - r)/(g + r - b)$	(Gitelson et al., 2002)

	Green Red Vegetation Index (GRVI)	$GRVI = (g - r)/(g + r)$	(Tucker, 1979)
	Modified Green Red Vegetation Index (MGRVI)	$MGRVI = (g^2 - r^2)/(g^2 + r^2)$	(Bendig et al., 2015)
	Color Index of Vegetation Extraction (CIVE)	$CIVE = 0.441 r - 0.881 g + 0.385 b + 18.78745$	(Machefer et al., 2020)
	woeExcess Green Index (EXG)	$EXG = 2 g - b - r$	(D. M. Woebbecke et al., 1995)
	Green Leaf Area Index (GLA)	$GLA = (2G - B - R)/(2G + B + R)$	(Louhaichi et al., 2001)
Texture	mean	Average red reflectance in a neighborhood	(Su & Gibeaut, 2017)
	variance	Variation in red reflectance in a neighborhood	

	homogeneity	The degree of similarity in pixel reflectance values within a region of an image	
	contrast	The difference between the darkest and lightest parts of the reflectance from each band.	
	dissimilarity	Similar to contrast but linearly weighted. Measures how different pixel pairs are.	
	entropy	Measures randomness or disorder in the reflectance of each image	
	second_momentum	Measures uniformity or energy. High when pixel pairs are similar.	
	correlation	Measures how correlated a pixel is with its neighbor. Sensitive to linear dependencies.	
Point cloud based	CHM	Seedling height from based on the seedling to the canopy top	(LaRue et al., 2019)
	vertCV	Coefficient of variation of height	

variables		values for all points	
	Q25	Average gap fraction in each 1 m thick horizontal slices clipped from bare ground to canopy top	
	Q50	Leaf area density within each canopy volume.	
	Q75	Vegetation area index. This is the sum of leaf area density values of each 1 m slice	

Classification Algorithms

We implemented three major classification algorithms in the R statistical program to first investigate the impact of algorithms for seedling detection from other vegetated classes. Across all three methods implemented, random forest provided the highest results and was used for further processing. Below section describes the parameterization of each algorithm we implemented. To allow reproducibility, a fixed set of seeds was provided (seeds = seeds), and pre-specified fold indices (index = folds) ensure the same data partitions are used across models. Additionally, predictions on the held-out folds are saved (savePredictions = TRUE) for further analysis or ensemble methods.

Random forest (RF)

A Random Forest classifier is trained using the `train()` function with the `method = "rf"`, which internally uses the `r randomForest` package. The response variable is vegetation class, and the model is trained using all predictors (vegetation indices, texture variables and point cloud based variables). The importance was set to “TRUE” to enable the calculation of variable importance scores, allowing for later interpretation of which input features (e.g., spectral bands, indices) most influence classification performance. Model training is executed in parallel (`allowParallel = TRUE`) to improve computational efficiency. A grid of candidate values for `mtry` (the number of predictors randomly sampled at each decision tree split) was defined using `tuneGrid`. Values tested include 2, 3, 4, 5, and 8, 12, allowing the algorithm to find the optimal level of predictor randomness for the ensemble.

A cross-validation procedure is defined using the `trainControl()` function ensuring consistent and robust model evaluation across classifiers. Cross-validation and metric computation are controlled by the `ctrl` object defined earlier. Specifically, k-fold cross-validation (`method = "cv"`) was used, with the number of folds defined by `n_folds` (`n_folds = 10`). A grid search strategy (`search = "grid"`) was employed to systematically test combinations of tuning parameters.

The `multiClassSummary` function was used to compute performance metrics tailored for multi-class classification problems, such as overall accuracy, kappa statistics, sensitivity, specificity, and others. Class probabilities were calculated (`classProbs = TRUE`) to enable probabilistic performance metrics or AUC curves, where applicable.

Support Vector Machine (SVM)

A linear Support Vector Machine (SVM) was trained using the `svmLinear3` method, which interfaces with the `LiblineaR` R package and supports both L1 and L2 regularization losses. The response variable was vegetation class while vegetation indices, texture variables, and height metrics were used as predictor variables.

A parameter grid is defined with two hyperparameters:

1. `cost`: the regularization parameter (C), controlling the trade-off between maximizing the margin and minimizing classification error. Values tested were 0.2, 0.5, and 1.
2. `Loss`: the type of loss function used during optimization. Two types were considered: L1 loss, which promotes sparsity and may ignore less informative features, and L2 loss, which usually generalizes better by penalizing large coefficients more smoothly.

Parallel computation was enabled (`allowParallel = TRUE`), and cross-validation procedures were managed using the `ctrl` object defined earlier.

Neural Network (nnet)

A forward neural network classifier was trained using the `nnet` method. The response variable was vegetation class, and vegetation indices, texture variables, and height metrics were used as predictors. A grid of hyperparameters was specified using `expand.grid()`:

1. `size` refers to the number of neurons in the hidden layer, with tested values of 5, 10, and 15.
2. `decay` represents the weight decay parameter (L2 regularization) to prevent overfitting, with tested values of 0.001, 0.01, and 0.1.

Training was limited to a maximum of 1000 iterations (`maxit = 1000`) to ensure convergence.

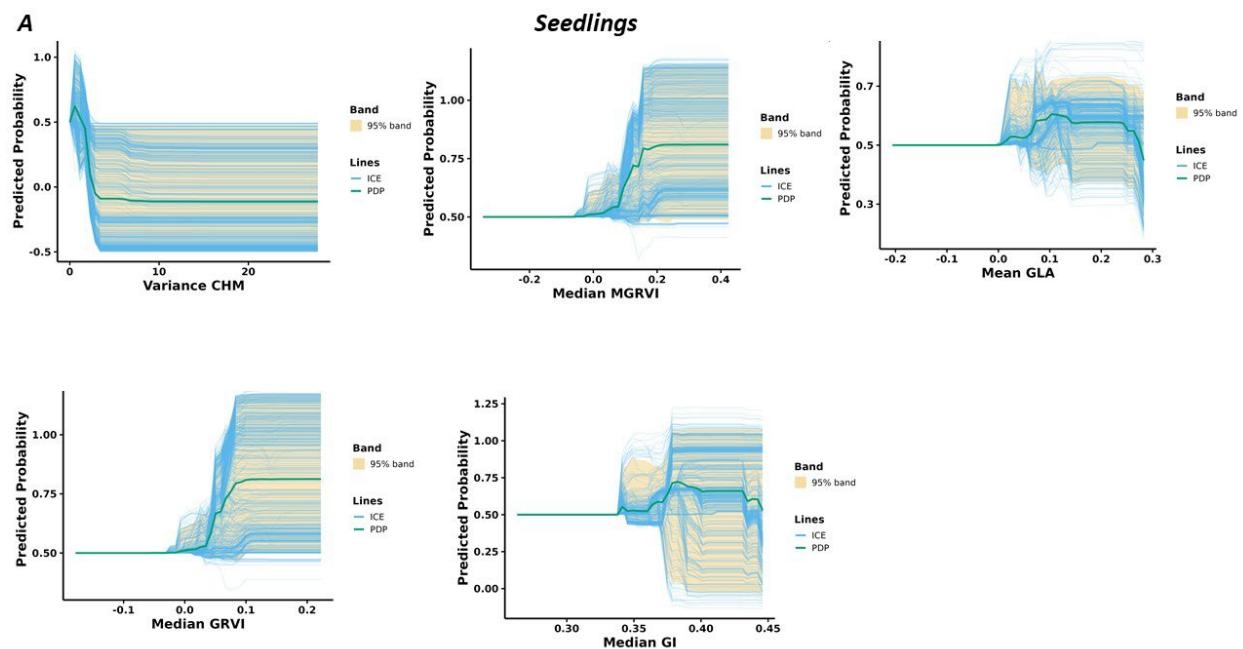
Although `importance = TRUE` was included, the `nnet` implementation in `caret` does not natively

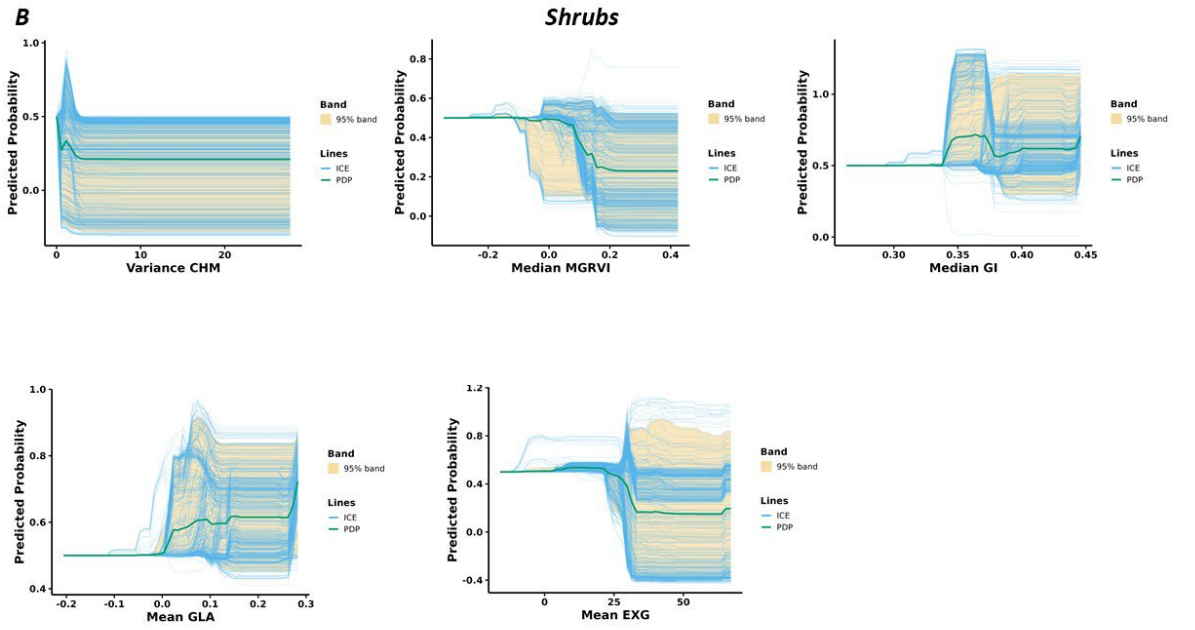
1 return variable importance unless extended with external tools. Nevertheless, the parameter is
2 retained for consistency.

3 Parallel execution was enabled (allowParallel = TRUE), and all model tuning and validation
4 settings were governed by the ctrl object.

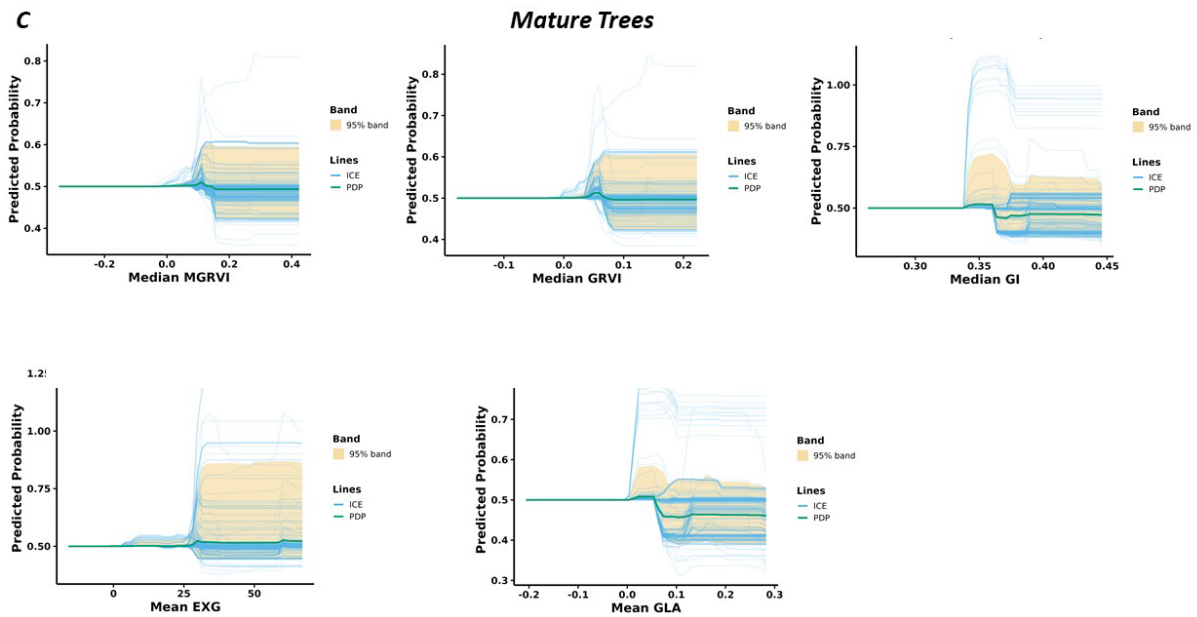
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6 **PDP and ICE plots for all variables of all classes**



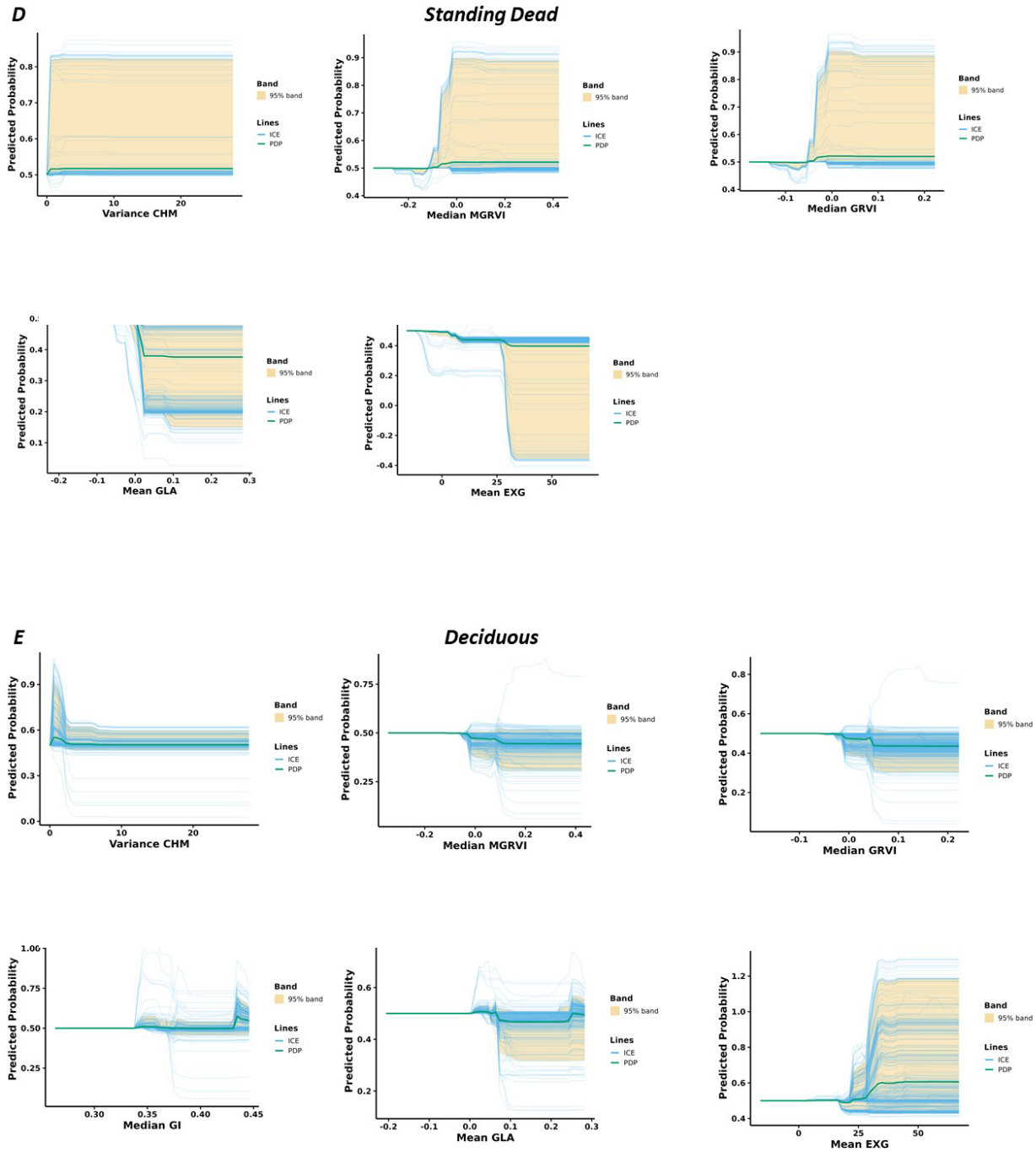


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Supplementary Figure 1: Partial dependence (PDP) and individual conditional expectation (ICE) plots showing the marginal effects of key predictor variables on the class probabilities. Each plot visualizes how changes in a selected predictor influence the predicted probability of assigning a

pixel to the corresponding class within a Random Forest classification model. Shaded areas represent the 95% confidence band around the PDP (green line), while blue lines show ICE curves for individual observations, highlighting variability in model response across the feature space.

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